*Objective of studying chemical reaction → to maximize formation of desired products.

* Various metrics are used to quantify effectiveness of a reaction to make products.

Ex: Consider a reaction: A (reactant) → B (desired product) + C (undesired products) occurring in a vessel called as Reactor.

1.Conversion = (amount of A consumed) / (amount of A fed to a reactor)

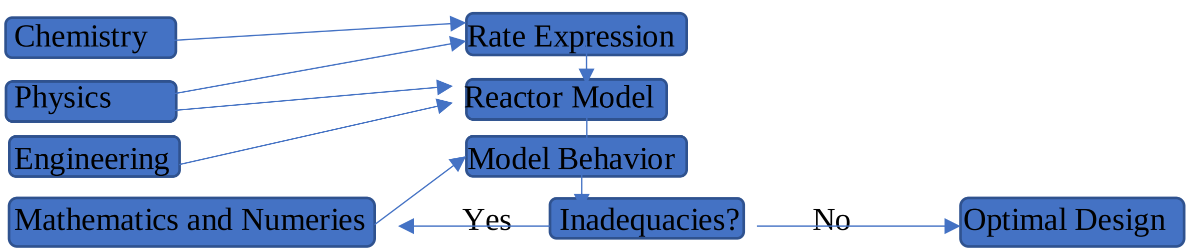
2.Yield = (amount of B produced) / (max amount of B that can be produced)

Yield is 100% when there is no undesired products (side reaction)

3.Seletivity = (amount of B produced) / (amount of C produced)

= (amount of B produced) / (total amount of A consumed)

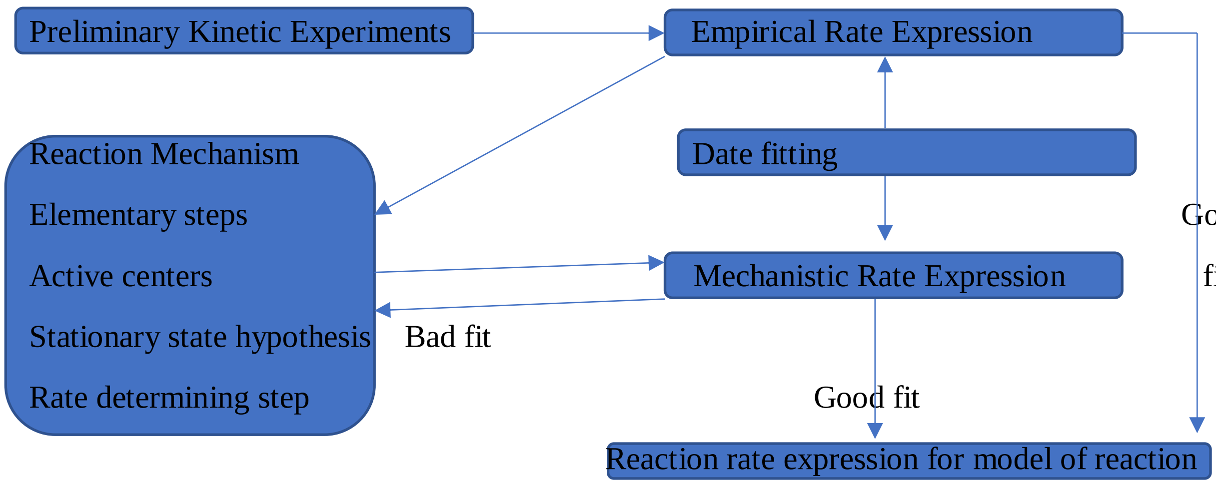
*How do we maximize conversion yield and selectivity of a desired product? We need to analyse chemical reactor. → Chemical Reactor Analysis (Ruthoford Aris : 1964)



*Optimal design of a reactor provides optimal dimension and operating condition for maximum conversion, yield, and selectivity of a desired product.

*Determining rate expression is crucial for optimal design of chemical reactor.

*Determining Rate Expression



***Kinematics of Chemical Reaction**

*Kinematics provide information regarding change in the molar composition of the reaction mixture, without concerning the rate of such change.

*In absence of transport effects, the change in molar composition is proportional to the stoichiometric coefficients.

*Consider a first order reaction: A → B

*1 mole of A is consumed for every mole of B produced.

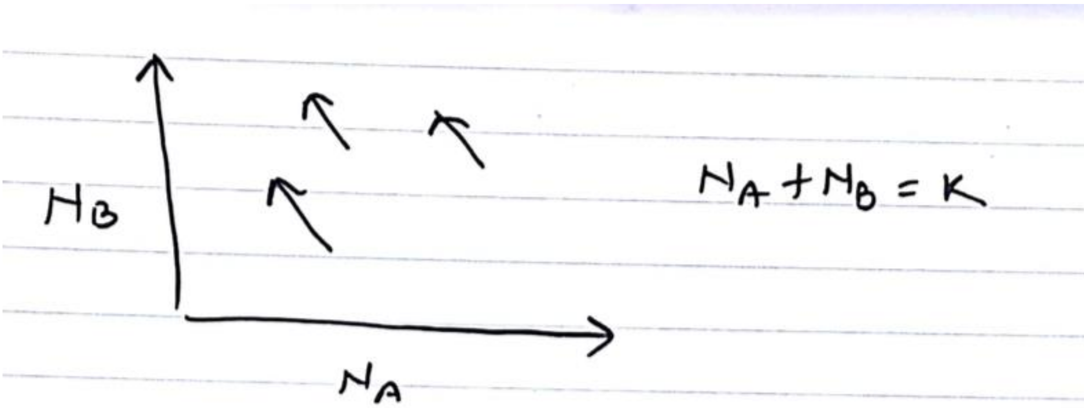
*the change in the moles of A (N_A) and the moles of B (N_B) in time dt are related as follow

$$-\frac{dN_A}{dt} = \frac{dN_B}{dt}$$

*the time interval dt is arbitray $-\frac{dN_A}{1} = \frac{dN_B}{1}$ (1)

*it is convenient to write A→B as B-A =0, such that the stoichiometric coefficient of product B is +1 and reactant A is -1.

*On (N_A, N_B) plane, equation (1) represents a course of reaction is a straight line of slope -1.



*In general, a chemical reaction involving s species (A_1, A_2, A_3 …… A_s) with stoichiometric coefficients α_1, α_2, α_3,…α_s can be represented as

$\alpha_1 A_1 + \alpha_2 A_2 + \ldots\ldots.. + \alpha_s A_s = \sum_{j=1}^{s} \alpha_j A_j = 0$ (2)

where $\alpha_j > 0$ for product, < 0 for reactant.

*From stoichiometry, we have: $\frac{d_{N_1}}{\alpha_1} = \frac{d_{N_2}}{\alpha_2} = \frac{d_{N_3}}{\alpha_3} = \cdots \ldots \ldots . = \frac{d_{N_s}}{\alpha_s} = dx$ (3) , which is a positive number regardless of species, and knows as the differential molar extend of reaction.

$dN_j = \alpha_j dx$ (4)

*In the foregoing, we said nothing about the volume V of the mixture, which may or may not remain constant during the course of reaction.

*For a fixed number of moles, varying the volume of reaction will affect the concentration and hence the rate of reaction.

*For a small volume dV and number of moles dN_j of A_j in it, the concentration can be defined as $c_j = \lim_{dV\to 0} \frac{dN_j}{dV}$ (5).

Mass concentration can be defined as $p_j = \lim_{dV\to 0} \frac{dM_j}{dV}$ (6) where $M_j = m_j N_j$. (m_j: molecular weight)

*Equation (4), for molar change in volume dV is d(∂N$_j$) = $\alpha_j d(\partial x)$

Molar extend per unit volume can defined as $\xi = \lim_{\partial V \to 0} \frac{\partial x}{\partial V}$ (7)

Likewise, molar extent per unit mass can be defined as $\xi'' = \lim_{\partial m \to 0} \frac{\partial x}{\partial M}$ where M=$\sum_{j=1}^{s} M_j$, which is useful for dealing with reactions which occur with change in volume.

*c_j, p_j, ξ, ξ'' are local quantities varying with position and time in a reactor.

*Local reaction rates can be defined as $\partial R_j = \frac{d\,(\partial N_j)}{dt} = \alpha_j \frac{d(\partial x)}{dt} \rightarrow \partial R = \frac{\partial R_j}{\alpha_j} = \frac{d(\partial x)}{dt}$ (9)

*Rate of formation of A$_j$ per unit volume $r_j = \lim_{\partial V \to 0} \left(\frac{\partial R_j}{\partial V}\right)$ from which intrinsic rate of reaction can be defined as $\gamma = \frac{\gamma_j}{\alpha_j} = \lim_{\partial V \to 0} \left(\frac{\partial R}{\partial V}\right)$ (10)

$$\gamma'' = \lim_{\partial M \to 0} \left(\frac{\partial R}{\partial M}\right) = \lim_{\partial V \to 0} \left[\left(\frac{\partial R}{\partial V}\right) / \left(\frac{\partial M}{\partial V}\right)\right] = \frac{\gamma}{p}$$

***Dynamic of Chemical Reactions**

*Dynamics of Chemical Reaction provide the change in mole of reaction mixture w.r.t time.

*Representing dynamics require detailed consideration of transport in a reactor.

*There are two types of reactors: (1) well-mixed batch reactor (2) continuous flow reactor

Well-mixed Batch Reactor (case of single reaction)

*Consider a general reaction $\sum_{j=1}^{s} \alpha_j A_j = 0$ in a well-mixed batch reactor of volume V containing N_j moles of A_j such that $dN_j = \alpha_j dx$ (12)

Quiz 2:

Consider a first order reaction involving reactant A and product B of initial molar composition N_{Ao}, N_{Bo} respectively. Find the maximum value of molar extent (X) when (1) reaction is irreversible. (2) reaction is reversible with equilibrium constant $K = \frac{C_B}{C_A}$. Assumed fixed volume.

Ans:

(1) Irreversible reaction: $X = \frac{N_A - N_{Ao}}{-1}$. The lowest possible value of $N_A = 0 \rightarrow$ max $X = N_{Ao}$.
(2) Reversible reaction. The maximum possible X can be realized when reaction attains equilibrium. Assume x moles of A needs to be consumed to attain equilibrium.

$$\rightarrow N_A - N_{Ao} = -x$$

$$K = \frac{C_B}{C_A} = \frac{\frac{N_{Bo}+x}{V}}{\frac{N_{Ao}-x}{V}} \rightarrow x = \frac{N_{Ao}K - N_{Bo}}{1+K}.$$

Also, $X = x = \frac{N_{Ao}K - N_{Bo}}{1+K}$ which can be integrated to $N_j = N_{jo} + \alpha_j X$ (13) where N_{jo} is the initial number of moles of A_j

*This eqn is the reaction invariant where molar composition must lie along the line represented by $\frac{N_1 - N_{1,o}}{\alpha_1} = \frac{N_2 - N_{2,o}}{\alpha_2} = \frac{N_3 - N_{3,o}}{\alpha_3} = \cdots \ldots \ldots . = \frac{N_s - N_{s,o}}{\alpha_s}$ (14)

*If we multiply m_j with eqn (13), we get $M_j = M_{j,o} + \alpha_j m_j X$. Sum over all species

$$\sum_{j=1}^{s} M_j = \sum_{j=1}^{s} M_{j,o} + \sum_{j=1}^{s} \alpha_j m_j X$$

$$M_{\text{total mass}} = M_{\text{total mass}} + X \sum_{j=1}^{s} \alpha_j m_j \rightarrow \boxed{\sum_{j=1}^{s} \alpha_j m_j = 0} \quad (16)$$

*In term of extent per unit volume , we can write dynamic balance as follow

$$\frac{dN_j}{dt} = \alpha_j \frac{dX}{dt} \rightarrow \frac{d(VC_j)}{dt} = \alpha_j rV \rightarrow c_j \frac{dV}{dt} + V\frac{dc_j}{dt} = \alpha_j rV \rightarrow \frac{dc_j}{dt} = \alpha_j r - c_j \frac{d\ln V}{dt} \quad (17)$$

*Volume change at constant P and T can be obtained from the equation of state. Total volume of reaction mixture can be related to partial molar volume as V = $\sum_{j=1}^{s} N_j V_j$ (V_j: partial molar volume).

Differentiate: $\frac{dV}{dt} = \sum_j \frac{dN_j}{dt} V_j + \sum_j \frac{dV_j}{dt} N_j$ and $\frac{dV}{dt} = \sum_{j=1}^{s} \alpha_j V_j R \rightarrow \frac{dlnV}{dt} = \sum_{j=1}^{s} \alpha_j V_j r$ (18)

Substitute (18) in (17): $\frac{dc_j}{dt} = \alpha_j r - c_j \alpha_j V_j R$ (19)

If the reactor volume remains constant $\frac{dc_j}{dt} = \alpha_j r$ (20)

*In term of ξ, we may write: from (9),

$\frac{dX}{dt} = R = Vr \rightarrow \frac{d(V\xi)}{dt} = Vr = r - \xi \frac{dlnV}{dt} = r - \xi \sum_{j=1}^{s} \alpha_j V_j r$

For a constant V, $\frac{d\xi}{dt} = r$ (21)

* In term of ξ'', we may write: from (15),

$M_j = M_{j,o} + \alpha_j m_j X \rightarrow \frac{dM_j}{dt} = \alpha_j m_j \frac{dX}{dt} = \alpha_j m_j M \frac{d\xi''}{dt}$ (22)

Also, $\frac{dM_j}{dt} = \alpha_j m_j \frac{dX}{dt} = \alpha_j m_j R = \alpha_j m_j Vr = \alpha_j m_j Vpr'' = \alpha_j m_j Mr''$ (23)

Compare (22) and (23), $\frac{d\xi''}{dt} = r''$ (24) (note that there is no effect of volume change on ξ'')

*The change in mass of individual species can be obtained from (23)

$\frac{dM_j}{dt} = \alpha_j m_j Mr'' \rightarrow \frac{d(M_j/M)}{dt} = \alpha_j m_j r''$

Let $g_j = (M_j/M)$ be the mass fraction. $\frac{dg_j}{dt} = \alpha_j m_j r''$ (24)

****Well-mixed Batch Reactor (case of multiple reactions)**

*Suppose there are m independent reaction taking place in a reaction.

$\alpha_{11} A_1 + \alpha_{12} A_2 + \ldots\ldots.. + \alpha_{1s} A_s = 0$

$\alpha_{21} A_1 + \alpha_{22} A_2 + \ldots\ldots.. + \alpha_{2s} A_s = 0$

….

$\alpha_{m1} A_1 + \alpha_{m2} A_2 + \ldots\ldots.+ \alpha_{ms} A_s = 0$

You can write reactions as

$\sum_{j=1}^{s} \alpha_{ij} A_j = 0$, i=1,2,…m

Consider a stoichiometric matrix $\underline{\underline{\alpha}} = \begin{bmatrix} \alpha 11 & \alpha 12 & \ldots & \alpha 1s \\ \alpha 21 & \alpha 22 & \cdots & \alpha 2s \\ \vdots & \vdots & \vdots & \vdots \\ \alpha m1 & \alpha m2 & \cdots & \alpha ms \end{bmatrix}$, $\underline{\underline{\mathbf{A}}} = \begin{bmatrix} A1 \\ A2 \\ \vdots \\ As \end{bmatrix}$

Number of independent chemical reactions is equal to the rank of $\underline{\underline{\alpha}}$.

*The reaction invariant for the multiple reaction case can be determined as

$N_j = N_{j,o} + \sum_{i=1}^{m} \alpha_{ij} X_i$, j = 1,2,…,s $\rightarrow \underline{N} = \underline{N_o} + \underline{\underline{\alpha}}^T \underline{X}$

$\rightarrow \underline{N} - \underline{N_o} = \underline{\underline{\alpha}}^T \underline{X}$ (26) or $(\underline{\underline{\alpha}}\, \underline{\underline{\alpha}}^T)^{-1} \alpha\, [\underline{N} - \underline{N_o}] = \underline{X}$ (27)

For a fixed molar extent $\underline{X}$, the reaction invariants (27) are m planes in s-dimensional space.

*The reaction invariant is a domain of (s-m) dimension in s dimensional space. Reaction invariants are very useful for making economic evaluation of a chemical reaction without kinetics.

Dynamics of a batch reactor can be obtained as follows:

$\frac{dN_j}{dt} = \sum_{i=1}^{m} \alpha_{ij} \frac{dX_j}{dt} \rightarrow \frac{d(VC_j)}{dt} = V \sum_{i=1}^{m} \alpha_{ij}\, r_i \rightarrow \frac{dC_j}{dt} = \sum_{i=1}^{m} \alpha_{ij}\, r_i - C_j \frac{dlnV}{dt}$ (28)

In a matrix form: $\underline{\frac{dC}{dt}} = \underline{\underline{\alpha}}^T\, \underline{r} - \underline{c} \frac{dlnV}{dt}$ (29)

In term of ξ, we may write: from (9),

$\frac{dX_j}{dt} = R_i = V r_i \rightarrow \frac{d(V\xi_j)}{dt} = V r_i \rightarrow \frac{d\xi_j}{dt} = r_i - \xi_j \frac{dlnV}{dt}$ (30)

*Volume change at constant T and P can be written as $\frac{dlnV}{dt} = \sum_{i=1}^{m} r_i \sum_{j=1}^{s} \alpha_{ij} V_j$ (31)

*Pressure change at constant T and V can be obtained as

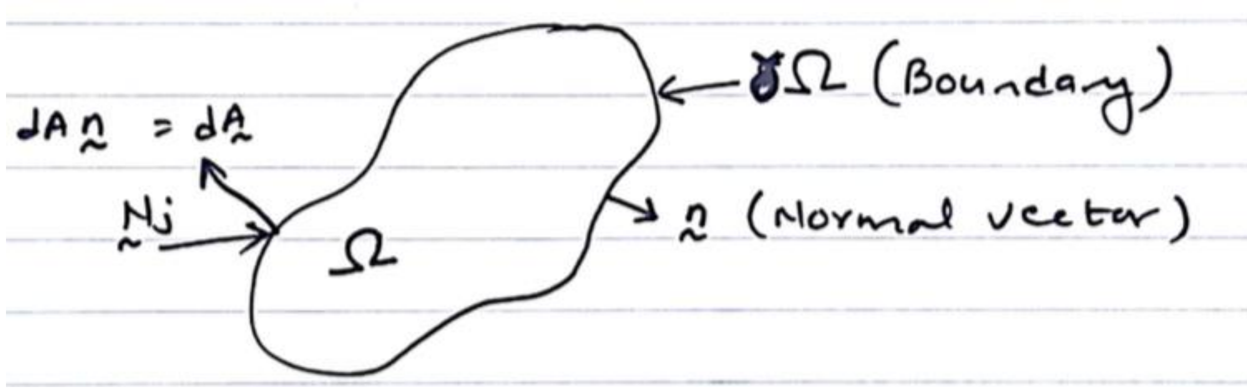
$\frac{dV}{dt} = 0 = \left(\frac{\partial V}{\partial P}\right)_{T,Nj} \frac{dP}{dt} + \sum_{j=1}^{s} V_j \frac{dN_j}{dt} \rightarrow \frac{dP}{dt} = \frac{-\sum_{j=1}^{s} V_j \sum_{i=1}^{m} \alpha_{ij} r_i}{\left(\frac{\partial lnV}{\partial P}\right)_{T,Nj}}$ (32)

In term of ξ'', $\frac{d\xi''}{dt} = r_i''$, i=1,2,…,m (33)

*****Continuous flow reactor**

*Continuum equations for reaction with transport in open systems.

*Consider a control volume (Ω) in a open system containing reaction mixture. Domain Ω is fixed, however, the boundary ∂Ω can exchange species with the surrounding.



*Mole balance in the control volume can be written as :

Rate Accumulation = rate of exchange at boundary + rate of production in domain

$$\frac{d}{dt}\int_{\Omega} C_j dV = -\oint_{\partial\Omega} \boldsymbol{N_j}.d\boldsymbol{A} + \int_{\Omega} r_j dV$$

Volume integral for total moles surface integral volume integral (34)

*The first term on right is the surface intergral of the dot product of flux (**N**$\mathbf{N_j}$) and area vector (d**A**). Molar flux $\mathbf{N_j} \equiv \frac{\boldsymbol{moles\ of\ A_j}}{\boldsymbol{area\ x\ time}}$

*For a positive accumulation (left), the moles of A_j must be entering through ∂Ω. Therefore, $\mathbf{N_j}$ pointing inward.

*The area d**A** is the normal vector times the differential area dA at the base of normal vector. d**A** and **n** both point outward to Ω.

*$-\boldsymbol{N_j}.\,d\boldsymbol{A}$ is the rate of moles of A_j entering the domain Ω. R_i is the net rate of production of species A_j per unit volume per time. $r_j = \sum_{i=1}^{m} \alpha_{ij} r_i$

*Apply Divergence theorem or Gauss theorem to eq 34 (first term on right)

$$\int_V (\nabla.\boldsymbol{F})dV = \oint_S (\boldsymbol{F}.\boldsymbol{n})dS \qquad (35)$$

Apply (35) to (34)

$$\frac{d}{dt}\int_\Omega C_j dV = -\int_\Omega (\nabla.\boldsymbol{N_j})dV + \int_\Omega \boldsymbol{r_j} dV$$

$$\int_\Omega \left(\frac{\partial C_j}{\partial t} + \boldsymbol{\nabla}.\boldsymbol{N_j} - \boldsymbol{r_j}\right) dV = 0$$

This equation is applicable to any size of the control volume Ω.

*The integral must be zero

$$\frac{\partial C_j}{\partial t} + \boldsymbol{\nabla}.\boldsymbol{N_j} = \boldsymbol{r_j} = \sum_{i=1}^{m} \alpha_{ij} r_i \qquad (36)$$

This is a continuing equation for reacting species.

****Flux of Species**

*For a moving species, the molar/mass flux is defined as moles/mass of species moving per unit time per unit area perpendicular to the direction of motion.

N_j ≡ total molar flux of A_j

n_j ≡ mass flux of species A_j = $p_j v_j$ = density x velocity of A_j

= $p_j (v_j - v) + p_j v$ = relative motion + bulk motion

Sum over all species $\mathbf{n} = \sum_{j=1}^{s} \boldsymbol{n_j} = \sum_{j=1}^{s} \boldsymbol{p_j v_j}$ ≡ total mass flux

Divide by density of solution p = $\sum_i p_i$, $\mathbf{v} = \frac{\boldsymbol{n}}{p} = \sum_{j=1}^{s} g_j v_j$ ≡ mass averaged velocity (38)

(note that the net notion of the mixture is due to bulk motion and not the relative motion)

*Define: $\mathbf{j}_j = p_j (v_j - v)$ ≡ mass diffusive flux (due to relative motion)

$p_j \mathbf{V}$ ≡ mass convective flux (due to bulk motion)

$\boldsymbol{n}_j = \boldsymbol{j}_j + p_j \boldsymbol{V}$ (39)

Divide by molecular weight m_j

$\frac{1}{m_j} \boldsymbol{n}_j = \frac{1}{m_j} \boldsymbol{j}_j + \frac{p_j}{m_j} \boldsymbol{v}$ → $\mathbf{N_j} = \mathbf{J_j} + C_j\mathbf{V}$ (40)

$\mathbf{N_j}$: molar flux

$\mathbf{J_j}$: molar diffusive flux w.r.t mass averaged velocity

$C_j\mathbf{V}$: molar convective flux w.r.t mass-averaged velocity.

*It is convenient to express flux w.r.t mass-averaged velocity because the mass of reaction mixture does not change with time.

Substitute (40) in (36)

$$\frac{\partial C_j}{\partial t} + \boldsymbol{\nabla}.\boldsymbol{J_j} + \boldsymbol{\nabla}.\left(C_j \boldsymbol{V}\right) = \sum_{i=1}^{\boldsymbol{m}} \alpha_{\boldsymbol{ij}} r_i$$

$$\frac{\partial C_j}{\partial t} + \boldsymbol{V}.\boldsymbol{\nabla} C_j + C_j(\boldsymbol{\nabla}.\boldsymbol{V}) + \boldsymbol{\nabla}.\boldsymbol{J_j} = \sum_{\boldsymbol{i=1}}^{m} \alpha_{\boldsymbol{ij}} r_{\boldsymbol{i}}$$

$$\frac{DC_j}{Dt} + \boldsymbol{\nabla}.\boldsymbol{J_j} = \sum_{\boldsymbol{i=1}}^{m} \alpha_{\boldsymbol{ij}} r_{\boldsymbol{i}} - C_j(\boldsymbol{\nabla}.\boldsymbol{V}) \quad (41)$$

***Continuity Equation in term of mass**

*Mass balance in the control volume yields on equ similar to (36)

$$\frac{\partial p_j}{\partial t} + \boldsymbol{\nabla}.\boldsymbol{n_j} = \sum_{\boldsymbol{i=1}}^{m} \alpha_{\boldsymbol{ij}} r_{\boldsymbol{i}} m_{\boldsymbol{j}} \qquad (42)$$

Substitute eqn (39)

$$\frac{\partial p_j}{\partial t} + \boldsymbol{\nabla}.\boldsymbol{j_j} + \boldsymbol{\nabla}.(p_j \boldsymbol{V}) = \sum_{\boldsymbol{i=1}}^{m} \alpha_{\boldsymbol{ij}} r_{\boldsymbol{i}} m_{\boldsymbol{j}}$$

Sum over all species

$$\sum_{j=1}^{s} \frac{\partial p_j}{\partial t} + \sum_{j=1}^{s} \boldsymbol{\nabla}.\boldsymbol{j_j} + \sum_{j=1}^{s} \boldsymbol{\nabla}.\left(p_j \boldsymbol{V}\right) = \sum_{j=1}^{s} \sum_{\boldsymbol{i=1}}^{m} \alpha_{\boldsymbol{ij}} r_{\boldsymbol{i}} m_{\boldsymbol{j}}$$

$$\frac{\partial}{\partial t} \sum_{j=1}^{s} p_j + \boldsymbol{\nabla} \sum_{\boldsymbol{j=1}}^{\boldsymbol{s}} \boldsymbol{J_j} + \boldsymbol{\nabla}.\left(\boldsymbol{V} \sum_{\boldsymbol{j=1}}^{s} p_j\right) = \sum_{\boldsymbol{i=1}}^{\boldsymbol{m}} r_{\boldsymbol{i}} \sum_{\boldsymbol{j=1}}^{s} \alpha_{\boldsymbol{ij}} m_{\boldsymbol{j}}$$

$$\frac{\partial p}{\partial t} + \boldsymbol{\nabla}.(\mathbf{V}\mathrm{p}) = 0 \qquad (43)$$

This is continuity eqn for mass

$$\frac{\partial p}{\partial t} + \boldsymbol{V}.\boldsymbol{\nabla} p = -p\boldsymbol{\nabla}.\mathbf{V} \rightarrow \frac{Dp}{Dt} = -p\boldsymbol{\nabla}.\mathbf{V}$$

(note $p = 1/\nu$ is specific volume = vol/mass)

$$-\nu \frac{D}{Dt}\left(\frac{1}{\nu}\right) = \boldsymbol{\nabla}.\mathbf{V}$$

$$\frac{D}{Dt}\left(\nu \frac{1}{\nu}\right) = \nu \frac{D}{Dt}\left(\frac{1}{\nu}\right) + \frac{D\ ln\nu}{Dt} = 0$$

$$\boldsymbol{\nabla}.\mathbf{V} = \frac{D\ ln\nu}{Dt} \qquad (44)$$

Substitute (44) in (41), we get: Open System

$$\frac{DC_j}{Dt} + \boldsymbol{\nabla}.\boldsymbol{J_j} = \sum_{\boldsymbol{i=1}}^{m} \alpha_{\boldsymbol{ij}} r_{\boldsymbol{i}} - C_j \frac{D\ ln\nu}{Dt} \qquad (45)$$

This is the overall continuity equation in terms of specific volume.

Compare with mole balance equ (28) for batch reactor (closed system)

$$\frac{dC_j}{dt} = \sum_{i=1}^{m} \alpha_{ij}\, r_i - C_j \frac{dlnV}{dt}$$